Coca-Cola's Digital Marketing

Analysis of our digital marketing campaign using A/B testing, analytical tools and ROI

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Who is Coca-Cola?

- One of the leading beverage brands in the US and worldwide
- Their most popular drink "coke" is one of the main selling products
- They also own other beverage companies outside of soda

What is our data?

- Comprised a dataset consisting of marketing spend for every media channel
  - TV, radio, social media
- Categorized into influencer advertisement types
- Sales resulted after each spend

What is our Mission?

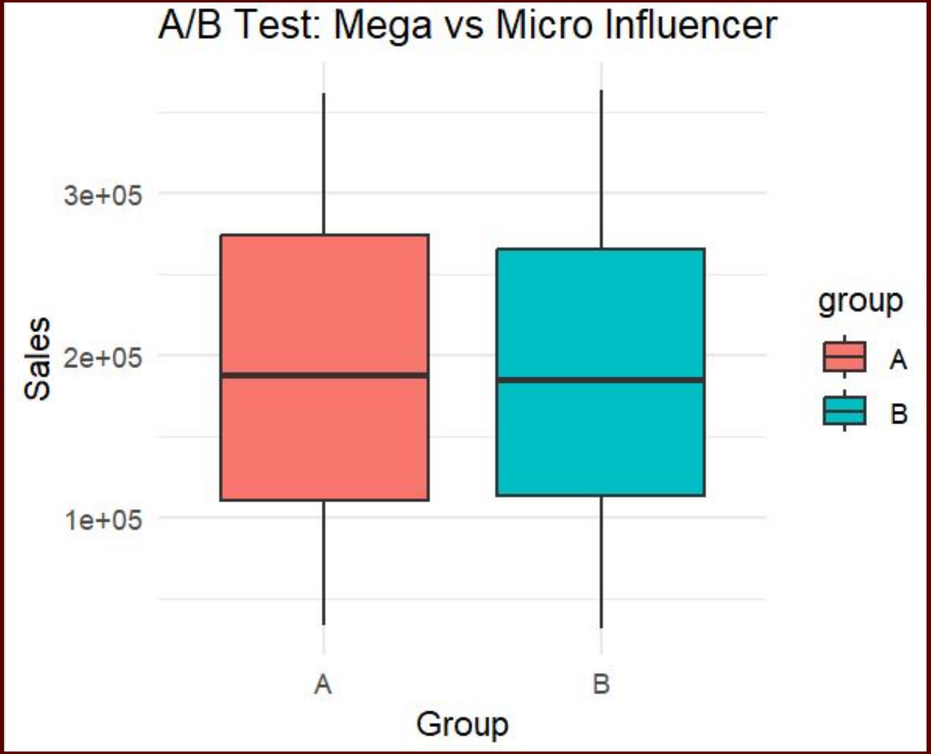
- We want to increase sales by 10% in the upcoming fiscal year
- Reduce marketing spend as much as possible
  - While maximizing sales
- Try to see what media platform has the best impact on sales

Our Tools & Methods

- R to conduct all of our data manipulations/visualizations
- We will be conducting A/B testing on influencer type
- Regression analysis on each media channel
- ROI analysis to see which media channel returns the most

A/B Test Results

- Used influencer type
    - A is Micro influencer (control)
    - B is Mega influencer (treatment)
- Found that the difference is not statistically significant
    - Different influencers do not have much impact on sales
- We can focus most on the media channels themselves instead of influencer type



p-value = 0.7644

Regression Analysis

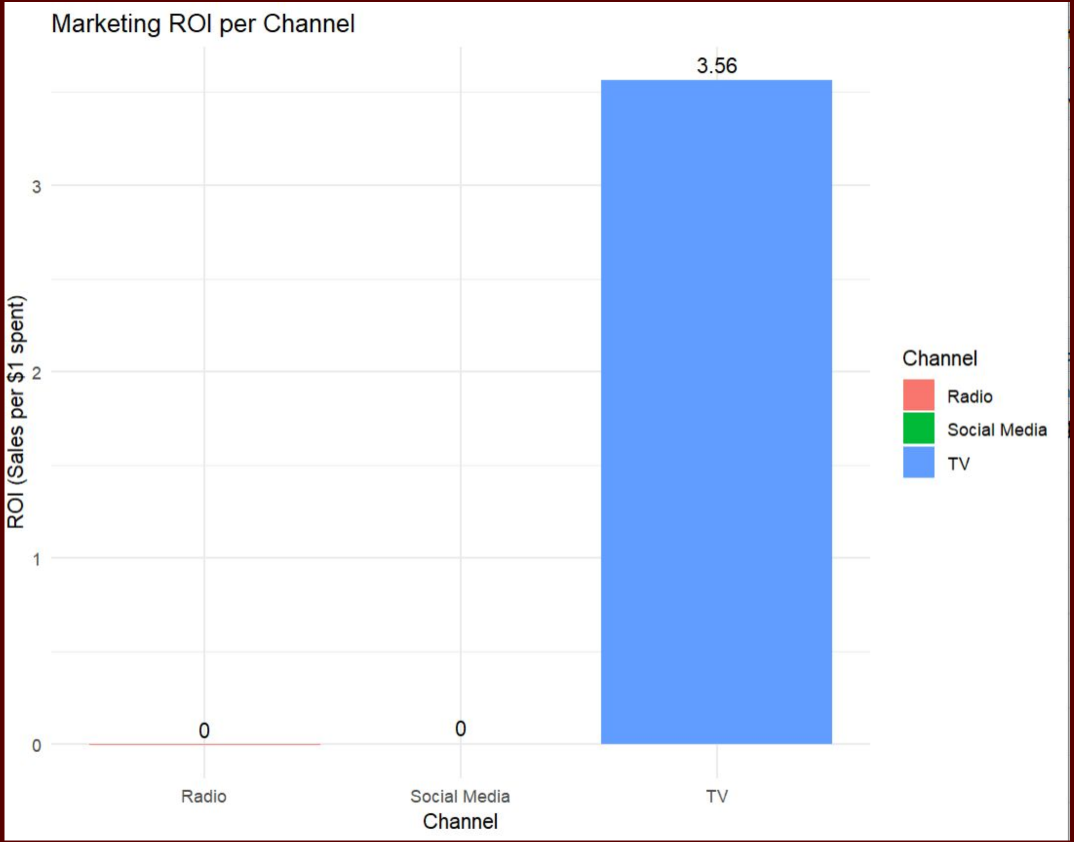
- TV is the only significant factor in sales (p-value < 0.05)
  - Other media channels had very little significance
- Highest coefficient for TV tells us that this media platform gives us the highest return per $1 spent on it
- Radio and Social Media show to have a very low return on sales

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	-1.033e+02	1.297e+02	-0.797	0.426	
tv	3.563e+00	3.391e-03	1050.472	<2e-16	***
influencerMega	1.160e+01	1.241e+02	0.094	0.926	
influencerMicro	-5.688e+01	1.242e+02	-0.458	0.647	
influencerNano	-7.390e+01	1.245e+02	-0.593	0.553	
social_media	4.464e-03	2.490e-02	0.179	0.858	
radio	-3.908e-03	9.785e-03	-0.399	0.690	

ROI Analysis

- Based on regression results, TV dominates when calculating the ROI
- For every $1 spent on TV advertising, sales increases by~$3.50

```
> coef(model)["tv"]
      tv
3.562565
>
> coef(model)["radio"]
       radio
-0.003907801
>
> coef(model)["social_media"]
social_media
 0.004464019
```



How to use this analysis?

- Now that we know TV advertising is the most significant variable, we can decide how to reach our desired outcome
- To reach our sales goal, we can calculate how much more we'll need in sales to reach this result
- Then we can use this to calculate how much more we'll need to spend for our most effective media channel

```
> current_sales <- mean(coke_data$sales)
> target_sales <- current_sales * 1.10  # 10% increase
> required_increase <- target_sales - current_sales
> required_increase
[1] 19241.33
```

```
> tv_needed <- required_increase / coef(model)["tv"]
> tv_needed
      tv
5400.978
```

Recommendations

- Focus on allocating budget towards TV advertisement
  - Consider lowering budget for radio and social media platforms
  - Reaching the budget requirement to increase our sales by 10%

Conclusion

- Influencer type has very little impact on sales
  - One type does not produce more sales than the other
- TV advertising is the most important media channel for predicting or increasing sales
  - It also returns the most, dominating over the other channels
  - For every $1 spent on TV advertising, ~$3.50 is returned
- We will need to increase sales by around ~$19,200 to reach our fiscal goal
  - By increasing the budget for TV advertisements by ~$5,400 we can reach our goal

Sources

- Marketing Spend Sales Impact Dataset CSV Download Free

Thank you!